

ELI — Agent Algorithms

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ELI Agent Algorithms — what each of the 15 agents actually computes

Updated 2026-09-30. Orchestrator runs for all CHAT modes; bus composed at S06. `_aggregate_confidence` and critic’s corroboration score both gained a query-relevance term.

Your framing is the right lens: **the DAG structures the reasoning steps; RAG is the engine that pulls the raw information during those steps.** ELI maps onto this cleanly —

- **DAG layer (structure):** the AgentOrchestrator 12-stage pipeline + the AgentBus topological layers (`eli/core/dag.py`) + the coding subtask DAG. This is the “search → summarize → check” skeleton.
- **RAG layer (raw information):** a *subset* of the agents are retrievers (`memory`, `knowledge_graph`, `file_code`) plus the orchestrator’s own HyDE→FAISS→RAG→rerank pipeline. These pull evidence.
- **Effectors / reporters:** the rest act (`system`, `plugin`) or report grounded runtime facts (`capability`, `introspection`, `frontier`, ...).

So there are **15 specialist agents in `_ALL_AGENTS`** plus the **AgentOrchestrator** that sequences them. SpecAgent instances and custom agents register on top.

Roles at a glance

Role	Agents
Retriever (RAG engine)	<code>memory</code> , <code>knowledge_graph</code> , <code>file_code</code>
Effector (acts on the world)	<code>system</code> , <code>plugin</code>
Grounded reporter (read-only facts)	<code>capability</code> , <code>voice</code> , <code>introspection</code> , <code>frontier</code> , <code>reflection</code> , <code>proactive</code> , <code>self_improvement</code> , <code>habit</code>
Planner (emits structure)	orchestrator (bus-level, descriptive) + the real AgentOrchestrator

Every agent shares the same skeleton: a **relevance gate** (cheap keyword/action test → skip with confidence=0 if irrelevant, so the bus stays fast) → its algorithm → an `AgentResult(ok, confidence, data)`. The

bus runs them in **topological layers** with **per-agent hard timeouts**, then `_aggregate_confidence` fuses them: `contribution = evidence_quality × evidence_density × learned_calibration × query_relevance`, single-agent-capped, corroboration-bonused. The query-relevance term (2026-09-29, distinct from the per-agent gate above) is `relevance_gate(term_overlap(user_input, representative_text))` — a fluent, evidence-dense result with nothing to do with what was actually asked no longer scores as highly as an on-topic one; CriticAgent's corroboration score gets the same discount (two sources agreeing with each other doesn't mean either is on-topic). Env `ELI_AGENT_BUS_RELEVANCE_GATE` disables it.

The 15 specialist algorithms

1. memory — hybrid dense+sparse retriever (RAG core) · 5.0s

- **Gate:** `_eli_memory_should_run` (explicit memory actions, memory markers, or ≥ 4 -word CHAT; blocks stale context bleeding into commands).
- **Algorithm:** `recall_memory = FAISS vector search first` (dense), with **FTS5/LIKE keyword search** as a supplement on cold/short queries (sparse); noise-filtered (drops ELI's own reflections/assistant-insights), importance-ordered. Plus `search_conversations` (FTS5), recent turns, session summaries.
- **Out:** `{results, conv_hits, memory_context}`. The primary RAG retriever.

2. system — deterministic action effector · 8.0s

- **Gate:** `action ∈ SYSTEM_ACTIONS` (121 OS and runtime actions); `LLM_ACTIONS` (`GENERATE_SCRIPT`, etc.) are **deferred** (never run inside the timed parallel phase — they'd time out + double-execute).
- **Algorithm:** delegates to `executor_enhanced.execute(action, args)` behind the security allowlist/sandbox. Confidence 0.92 ok / 0.20 fail. The hands.

3. habit — pattern-store reader · 3.0s

- **Gate:** habit/routine/schedule keywords or `OPEN_APP`.
- **Algorithm:** reads `habit_rules` (all) + `habit_events('app_launch', 14d)`; confidence by rule presence. (Frequency mining itself lives in the proactive daemon; the agent surfaces the rules + event volume.)

4. self_improvement — failure-cluster surfacer · 3.0s

- **Gate:** `self/improve/fix/learn` keywords or `SELF_*` actions.
- **Algorithm:** `analyze_failures(7d, clustered by occurrence_count)` + pending capability proposals. Feeds the self-upgrade loop (see `background_tasks.md`).

5. proactive — artifact reader · 3.0s

- **Gate:** `proactive/morning-report/insight` keywords or `PROACTIVE_*` actions.
- **Algorithm:** reads `artifacts/proactive/latest_{context,summary,action}.txt` - `PROACTIVE_STATUS`. Surfaces the background daemon's findings.

6. frontier — cross-system introspection report · 5.0s

- **Gate:** `FRONTIER_STATUS / ELI_IDENTITY_AUDIT` or “full system status”.
- **Algorithm:** `build_frontier_status_report / eli_identity_audit` — an import + wiring matrix across runtime/memory/self/proactive/image/world/labs. Deterministic whole-system audit.

7. plugin — plugin effector · 6.0s

- **Gate:** action ∈ PLUGIN_ACTIONS (ADD_EVENT, GET_WEATHER, LIST_EVENTS, REMOVE_EVENT).
- **Algorithm:** execute(action, args) into the plugin layer; silent skip otherwise.

8. capability — capability-manifest reader · 6.0s

- **Gate:** capability/awareness/“what can you do” or LIST_CAPABILITIES etc.
- **Algorithm:** execute(LIST_CAPABILITIES | AWARENESS_STATUS | CODE_CHANGES) → grounded capability surface (no confabulated feature lists).

9. voice — config reporter · 5.0s

- **Gate:** voice/tts/stt/mic keywords.
- **Algorithm:** reads ELI_TTS_ENGINE/ELI_PIPER_MODEL/ELI_MUTE + probes faster_whisper import. Reports, never emits audio (that’s the TTS router).

10. orchestrator (bus-level) — descriptive planner · 3.0s

- **Gate:** multi-step trigger groups (e.g. {document, save, open}) or GROUNDED_SYNTHESIS_ACTIONS.
- **Algorithm:** emits a structured plan dict (now also surfaced as the typed ExecutionPlan). **Excluded from confidence aggregation** (it plans, it isn’t evidence). The *executing* planner is the ReAct loop in AgentOrchestrator.

11. file_code — source-code RAG (grep retriever) · 4.0s

- **Gate:** grounded query or code/file/pipeline/db keywords.
- **Algorithm:** builds query-derived regex patterns, **greps a canonical files_map** of ELI’s own modules (agent_bus, engine, orchestrator, memory, router, ...), returns file:line: content snippets. Lets ELI answer “which file does X” from real source, not memory.

12. reflection — reflection-log reader · 4.0s

- **Gate:** reflection/pattern/noticed/insight keywords or grounded query.
- **Algorithm:** recent observations (≤8) + session summaries (≤3) → insight list. ELI’s “what I’ve noticed about how you use me.”

13. introspection — self-model reporter · 4.0s

- **Gate:** architecture/pipeline/“how do you work” or EXPLAIN_*_RUNTIME.
- **Algorithm:** IntrospectionAgent → get_pipeline() + get_memory_stats() + get_runtime() → grounded pipeline description. Powers honest self-explanation.

14. knowledge_graph — graph retriever (structured RAG) · 3.0s

- **Gate:** non-empty KG.
- **Algorithm:** context_for_prompt(query) over kg_entities/kg_relations (FTS5 fuzzy entity match + relation context). **DAG edge memory → knowledge_graph:** it now seeds the query with intent["_upstream"]["memory"] hits (topological layering), so the graph lookup is conditioned on what memory just surfaced.

15. critic — second-tier verifier (corroboration check) · 2.0s

- **Gate:** ≥2 grounded upstream sources — self-gates to nothing on trivial turns, so it never fires when there’s nothing to cross-check.

- **Algorithm:** the one bus agent that reasons over **other agents' output** (intent["_upstream"]) rather than the raw query — **deterministic, no LLM**. It pulls the representative text each retriever surfaced this turn, measures **pairwise term agreement**, and emits a **corroboration-vs-contradiction** signal. Corroboration alone isn't enough (2026-09-29): two sources can agree strongly with EACH OTHER while both being off-topic for what was actually asked, so on the non-contradiction branch, agreement is additionally discounted by how well the sources match the *query* (relevance_gate over their average term_overlap(user_input, text)), not just each other. `_representative_text` moved from a CriticAgent staticmethod to a module-level function, since `_aggregate_confidence` now needs it too — one owner. **DAG edges:** depends on memory, file_code, knowledge_graph and system, so it runs in a **downstream topological layer** and sees their results — giving the agent DAG a real second tier instead of a flat fan-out.

How the DAG sequences the RAG (your “search → summarize → check”)

1. **Route** → action + reasoning mode.
2. **AgentOrchestrator** runs for **all CHAT modes** (gradient depth) and for non-CHAT actions; for non-CHAT it dispatches the **bus** then runs a **ReAct loop** (tool → observe → decide → next tool) — the executing DAG of steps.
3. **AgentBus** builds the **agent DAG** (`_AGENT_DEPENDENCIES`), runs agents in **topological layers**, passing each layer's results downstream (memory → knowledge_graph). Retrievers pull evidence; reporters ground it; effectors act.
4. **Grounding/evidence layer** gates the synthesis (output_violates_evidence).
5. **Coding tasks** decompose into a **subtask DAG** (plan_graph) and heavy ones run on **background workers** — see dag.md + background_tasks.md.

That is the literal realisation of “DAG structures the autonomous reasoning steps while RAG pulls the raw information.”

Aspirational next steps (genuinely novel combinations)

These are the “never-before-put-together” moves the current architecture is one step away from:

1. **Bias/verification agent as an explicit DAG layer.** [done] **Shipped** — this is now the critic agent (#15 above): it depends on the retrievers (memory/system/ knowledge_graph) and runs *after* them as a topological layer — literally “search → summarize → **check for corroboration/contradiction**” — feeding the grounding signal. (Originally proposed here as future work; now built.)
2. **A web retriever agent** (WEB_SEARCH as a first-class RAG node) so external retrieval joins memory/KG/file_code in the same layer, with the verify layer cross-checking sources.
3. **Calibration-aware selection.** The bus already learns per-(agent,action) calibration; feed it back into *selection* so chronically-zero agents are dropped from the DAG for that action (closes the learned-trust/unlearned- selection gap noted in orchestration_and_agents.md).
4. **Bus↔coding bug-memory unification.** The coding engine's bug_memory and the agents' self_improvement failure log are two halves of one “what went wrong and how we fixed it” store — merging them gives ELI a single experiential memory across chat and code.
5. **Orchestrator stage-dict → typed ExecutionPlan DAG.** Fold the engine's rich stage plan into ExecutionPlan.steps as a true DAG so one plan type drives the whole pipeline end-to-end.

SpecAgent and the measurable-agent contract

A further agent class joins the fifteen built-ins. SpecAgent is not hand-written Python — it is an interpreter for an AgentSpec (see orchestration_and_agents.md), and its algorithm is deliberately small:

```
run(user_input, intent):
```

```

if not spec.should_run(user_input, intent.action):      # trigger gate
    return skipped                                     # costs one regex/substring
for cap in spec.permissions:                           # capability gate
    if not permissions.check(agent, cap).allowed:
        return denied
output = broker.infer(prompt=user_input,
                      system=spec.system_prompt,
                      max_tokens=spec.max_tokens,
                      temperature=spec.temperature,
                      background=True)
verdict = spec.evaluate(output)                        # the measure
return AgentResult(ok=verdict.ok,
                  confidence=verdict.score,
                  data={"content": output if verdict.ok else ""})

```

Three properties worth stating:

- **The trigger gate runs first and is cheap.** An untriggered turn costs a substring or regex test, not an inference call.
- **Confidence is earned, not asserted.** Every other agent picks its own confidence number. A SpecAgent's confidence *is* its score against its own declared success criteria — the fraction of checks its output actually passed.
- **Failing its own test yields no evidence.** `data["content"]` is emptied when the verdict fails, so a misbehaving custom agent cannot feed unchecked text into the bus's evidence fusion. This directly addresses the long-standing weakness that evidence density “only counts known keys”, by making the agent's own contract the arbiter.

Because a spec contains no code, SpecAgent bypasses the custom-agent trust chain entirely — there is nothing to hash, scan or approve.